

SQL Beginner Notes (Step by Step)

1. Create Database

Purpose

Used to create a new database.

Syntax

```
CREATE DATABASE database_name;
```

Example

```
CREATE DATABASE school_db;
```

Important Points

- Creates a new database
 - Database names should be meaningful
-

2. Use Database

Purpose

Select database for working.

Syntax

```
USE database_name;
```

Example

```
USE school_db;
```

3. Create Table

Purpose

Creates structure for storing data.

Syntax

```
CREATE TABLE table_name(  
  
column datatype,  
  
column datatype  
  
);
```

Example

```
CREATE TABLE students(  
  
id INT PRIMARY KEY,  
  
name VARCHAR(50)  
  
);
```

Important Points

- Table stores rows and columns
 - Choose correct data types
-

4. Constraints

Constraints are rules applied to columns.

NOT NULL

```
name VARCHAR(50) NOT NULL
```

Meaning:

Column cannot store NULL

UNIQUE

```
email VARCHAR(100) UNIQUE
```

Meaning:

Duplicate values not allowed

PRIMARY KEY

```
id INT PRIMARY KEY
```

Meaning:

Unique identity of row

CHECK

```
age INT CHECK(age >= 18)
```

Meaning:

Value must satisfy condition

DEFAULT

```
status VARCHAR(20) DEFAULT 'active'
```

Meaning:

Automatic value when none provided

5. Insert Data

Syntax

```
INSERT INTO table_name(columns)

VALUES(values);
```

Example

```
INSERT INTO students(

id,

name

)

VALUES(

1,

'Ram'

);
```

6. Select Data

All Columns

```
SELECT *

FROM students;
```

Specific Columns

```
SELECT name,age

FROM students;
```

7. WHERE Clause

Used for filtering.

```
SELECT *  
  
FROM students  
  
WHERE age>20;
```

8. UPDATE

Modify existing data.

```
UPDATE students  
  
SET age=25  
  
WHERE id=1;
```

Important:

```
Without WHERE → all rows update
```

9. DELETE

Remove rows.

```
DELETE FROM students  
  
WHERE id=1;
```

Important:

```
Without WHERE → all rows deleted
```

10. TIMESTAMP

Automatically stores time.

```
created_at TIMESTAMP  
  
DEFAULT CURRENT_TIMESTAMP
```

Auto update:

```
updated_at TIMESTAMP  
  
DEFAULT CURRENT_TIMESTAMP  
  
ON UPDATE CURRENT_TIMESTAMP
```

11. ORDER BY

Sorting rows.

```
SELECT *  
  
FROM students  
  
ORDER BY age DESC;
```

12. LIMIT

Restrict number of rows.

```
SELECT *  
  
FROM students  
  
LIMIT 2;
```

Common Beginner Mistakes

Forgetting WHERE in UPDATE

Forgetting WHERE in DELETE

Using wrong datatypes

Confusing PRIMARY KEY and UNIQUE

Inserting wrong values into columns